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Privacy-Preserving Federated Transformer Framework for Detecting Adversarial Prompts in Healthcare Chatbots

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***Abstract*—Prompt injection attacks is an important security problem in healthcare-specific large language model systems(LLMs). The injected prompt can be used to control the behavior of the model or to leak sensitive data. This vulnerability increases with centralized training pipelines that collect patient prompts on a single server, which leads to data leakage and model poisoning. In this work, we examine Federated Learning (FL) as a privacy-preserving and decentralized prompt injection detection defense for medical chatbots. Using the SPML Healthcare Chatbot Prompt Injection Dataset, we fine-tune a RoBERTa-base classifier in conditions of centralized, IID (Independent and Identically Distributed), federated and Non-IID (Non-Independent and Non-Identically Distributed) federated learning. The experimental results show that the federated model has comparable performance with centralized training, with accuracy of 0.957 and F1-score of 0.964 under IID conditions. The realistic case scenario of Non-IID client distribution was evaluated across 3 different client configurations, the model is able to maintain stable performance with an average performance accuracy of 0.956, precision of 0.942, recall of 0.987, F1-Score of 0.964 and AUC of 0.994. Moreover, the federated setup shows consistent convergence between the communication rounds despite the data heterogeneity. These results suggest that transformer-based federated learning provides an effective and privacy-preserving solution for detecting prompt injection attacks in distributed healthcare conversational AI systems. This work contributes to Sustainable Development Goal (SDG) 3: Good Health and Wellbeing, SDG 9: Industry, Innovation and Infrastructure, and SDG 16: Peace, Justice and Strong Institutions by enhancing the security and privacy of AI-driven healthcare technologies.**

***Index Terms*—Prompt Injection Attacks (PIA), Federated Learning (FL), Healthcare Chatbots, RoBERTa, Natural Language Processing (NLP), AI Security, Sustainable Development Goals (SDG 3, SDG 9, SDG 16).**

I. Introduction

The progress in LLMs such as as GPT, BERT, and RoBERTa has contributed to NLP to a great extent which helped in understanding clinical note, to get medical assistance. The increasing dependence on prompt-based interaction has introduced a security risk known as PIAs. These attacks are done by inserting malicious instructions into user prompts to override model controls that leads to misdiagnosis, unsafe clinical advice or give out sensitive medical information, these are harmful threats to medical apps such as telemedicine systems, digital triage systems, and clinical chatbots [1], [2]. From the recent studies, LLM-based conversations can be exploited for social engineering and fraudulent activities, these strongly show the need to protect healthcare chatbots against adversarial prompt manipulation [3]. LLMs follow instructions, prompts are crafted in such a way that this trait is exploited[1]. Generative AI increases the risk of harmful content spreading across online platforms, which asserts the need to detect and prevent malicious prompts before they corrupt the sensitive healthcare data and information [4].The attack results in sharing protected health information (PHI) or giving untrustworthy medical advice, creating serious concerns for healthcare privacy regulations [2]. As conversational AI systems are widely used in clinical workflows they are prone to prompt injection attacks[1].

Existing defenses such as input sanitization and context isolation struggle to detect semantically complicated attacks that bypass the pattern-based detection[2]. Most of the methods are based on centralized training, where raw patient data are centralized on one server. This makes it easier to poison data, manipulate models, and leak sensitive clinical data, making healthcare systems more vulnerable to privacy and security breaches [5]. These risks are especially crucial in the medical environment, where regulations such as HIPAA and GDPR emphasize the need to adopt decentralized learning technique that will not collect sensitive data, ensuring that the model remains intact.

FL is one of the technologies that can resolve these issues by facilitating the joint training of models in a group of healthcare

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organizations without transmitting raw patient data. Every client conducts local training and transfers only the model parameters to aggregate it, enhancing privacy and minimizing risks [6], [7]. Previous work identifies FL as a good approach in healthcare applications and robust in the face of heterogeneous data distributions. Cross-domain research also indicates that FL can compete in sensitive fields and maintain data confidentiality at the same time.

In this paper, we explore how FL can be used to detect prompt injection attacks within the medical chatbot. A RoBERTa-base classifier is trained on SPML Healthcare Chatbot Prompt Injection Dataset [8] using centralized and federated training using a variety of client settings to assess robustness in realistic Non-IID settings. The paper focuses on performance consistency, privacy maintenance, minimized exposure in attack, and convergence stability in the situation of unequal client involvement.

II. Literature Review

A. Prompt Injection Attacks in Healthcare AI

PIAs utilize how LLMs interpret and execute naturallanguage instructions. LLMs can be manipulated to bypass system constraints, expose confidential data, or behave in unintended ways [1]. Advanced attacks, including mathematically encoded or function based prompts , can bypass simple keyword filtering and context isolation mechanisms. These show the limitations of input validation [2].

These attacks should not be taken lightly as strict privacy requirements and safe decision making is involved in healthcare sector. Multimodal medical vision language models [9] can be exploited to generate misleading diagnoses. This indicates that PIAs pose serious problems on high-stakes clinical settings.

B. Defenses Against Prompt Injection in Medical Systems

Techniques such as filtering, sanitization, and context separation [1], [2], provide security but they struggle with complex attacks in healthcare chatbots. Hybrid machine learning approaches can provide improved robustness and anomaly detection capabilities, which can also be relevant for detecting adversarial prompts in healthcare chatbots[10]. Most existing solutions rely on centralized approach where patient information is processed on a central server which increases the risk of information leakage, data poisoning, and unauthorized access to PHI.

C. Centralized Learning Vulnerabilities in Healthcare

Central systems makes use of the PHI for training and this information provides model updates. Centralized system can be attacked by poisoning attacks or to leak information [7], [11]. Such attacks not only affect technical systems but also the trust on such LLMs[12]. Even simple manipulation of the parameters can affect the performance or cause bias [7]. HIPAA and GDPR are some of the regulations that limit centralized management of sensitive information. New AI models focus on the decentralized and distributed learning in order to enhance security and scalability [6].

D. Federated Learning for Secure Healthcare NLP

FL provides a decentralized approach, where local models are trained at clients and the combined model updates are exchanged, which reduces PHI sharing and adversarial propagation. Light and distributed security solutions emphasize the significance of efficient and adaptive detection of the resourceconstrained environment [13]. FL is capable of reaching the performance of centralized learning and keeping privacy intact at the same time [6], [14].It enhances the resistance against poisoning and inference attacks, also efficient transformer updates are used to minimize the communication overheads[7], [11]. Balancing robustness, latency, and resource cost is an issue in distributed healthcare systems [15].

E. Research Gap

PIAs and FL are being studied individually, but little research has been done to understand and study how the two can be integrated into healthcare settings. Existing PIA defenses are based on centralized methods, and only few FL research pay attention to adversarial prompt detection. In this work, we propose a federated transformer-based system powered by the SPML Chatbot Prompt Injection Dataset [8]with a focus on balancing detection performance, privacy preservation, and healthcare regulatory requirements.

III. Proposed Methodology

A. System Overview

The proposed framework is to identify prompt injection attacks to healthcare-based LLM systems without requiring the centralized storage of sensitive clinical data. To achieve this, we combine FL with RoBERTa-based text classification model. Each participating healthcare institutions or client devices trains the model with their own medical prompts, so that the raw patient data never leaves the client side. Only model updates are shared with the central server, which helps reduce the risk of PHI exposure and related privacy issues.

This framework follows the basic principles of FL described in [6], [7], but adapts them for adversarial prompt detection in clinical dialogue systems. The main focus of the project is to have a strong detection performance while reducing the risk of data leakage, model poisoning , and privacy violations. This approach is suitable for healthcare settings where patient data confidentiality is highly important.

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B. Architectural Design

The proposed architecture consists of two layers.The first layer consists of participating client institutions. Each client has its own benign and adversarial prompt data and performs local fine-tuning of the RoBERTa model without sharing raw patient text or clinical conversations. The data never leaves the client side which address PHI-related privacy concerns and supports and obeys with healthcare privacy laws.

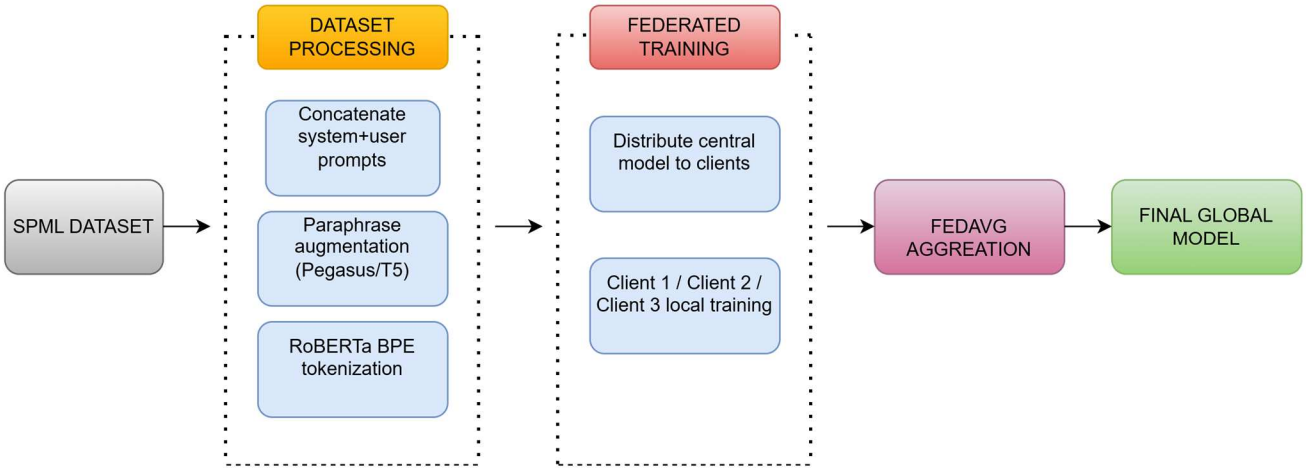


Fig. 1. Overall federated learning architecture for healthcare prompt injection detection.

The second layer is the federated server layer, which manages the global learning process. The server collects locally updated model parameters from each client and aggregates them using the standard FedAvg rule:

$$w_{t+1} = \sum_{i=1}^{K} \frac{n_i}{n} w_i^t \tag{1}$$

Here, w_i^t represents the local model parameters from client *i* at communication round *t*, n_i denotes the number of samples at that client, and *n* is the total number of samples across all participating clients. This weighted averaging strategy has proven effective in prior studies [16], [17]. Since only model updates are exchanged, the framework reduces the need for centralized storage of raw healthcare data.

C. Data Preprocessing Pipeline

For experimental evaluation, we make use of SPML Chatbot Prompt Injection Dataset [8], this dataset consist of system prompts, user prompts, and their corresponding labels. As the work focuses on healthcare related prompts , the dataset was filtered to retain only healthcare entries.

To preserve conversational semantics, the system and user prompts were combined into a single input sequence before tokenization. The merged text was then processed using the RoBERTa tokenizer based on Byte-Pair Encoding (BPE) [1]. This produces token IDs and attention masks, which are later converted into PyTorch tensors such as input_ids, attention_mask, and labels. This preprocessing step helps retain both contextual information and clinical intent, allowing the model to better distinguish normal healthcare conversations from adversarially manipulated prompts.

D. Model Configuration

We chose RoBERTa-base after initial experiments of comparison with DistilBERT-base-uncased and DeBERTa-v3-base. Although all three models demonstrated a similar level of performance under a centralized context, RoBERTa demonstrated higher levels of stability and robustness when trained with federated constraints. Each client fine-tunes the RoBERTa model locally using the following configuration:

- learning rate of 2×10^{-5},
- batch size of 8,
- three local training epochs per communication round,
- AdamW optimizer.

Once the clients have done their local updates, they transmit only the updated model weights to the federated server. The server gathers together these updates and sends the new global model for the next communication round. This iterative process continues until the global model gets to a stable state.

E. Evaluation Metrics

We evaluate model performance using standard classification metrics, including Accuracy, Precision, Recall, and F1score:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = 2\times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Here, *TP*, *FP*, and *FN* correspond to true positives, false positives, and false negatives when identifying malicious prompts. Because in real-world healthcare data are not necessarily balanced across healthcare institutions, we also

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analyze the stability of the model under both IID and Non-IID client partitions, following established evaluation practices [7], [11].

This helps to understand how effectively the framework can handle distribution differences that commonly arise in multihospital and distributed healthcare environment.

IV. Experimental Setup

A. Dataset Preparation

All experiments were done using the healthcare subset of the SPML Prompt Injection Dataset [8], which has both benign and adversarial prompts designed to simulate prompt injection attacks in medical chatbots. The system prompts and user prompts were combined before preprocessing. Since the dataset was imbalanced, paraphrase-based data augmentation was applied using a Pegasus-style transformer model (Vamsi/T5 Paraphrase Paws).Beam search (num beam=3) and temperature tuning were used to generate semantically similar but diverse prompt variations. The generated samples were then filtered to have only natural and unique prompts.After preprocessing and augmentation, the final dataset was 7,243 samples of which 2,949 were benign and 4,294 were malicious.

B. Client Data Partitioning

The dataset was allocated to three clients in order to simulate distributed healthcare environments. The IID setting randomly shuffled and split samples such that a client received an equal amount of benign and malicious prompts. In the case of Non-IID experiments, data heterogeneity was added through Dirichlet partitioning with concentration parameters of α =1.0, 0.5, and 0.3, which are mild, moderate and severe skew, respectively. The federated training process consisted of four communication rounds where all clients participated in each round. Only the model updates were exchanged with the central server, and the results were averaged over three runs.

C. Training Setup

The federated experiments used three clients performing local fine-tuning followed by FedAvg aggregation at the server. RoBERTa-base served as the backbone model. The same tokenization and preprocessing rules were applied across all clients.

D. Dataset partitioning

In order to decrease sensitivity to a single client partition, the experiment was run several times on different random NonIID client splits created under varying Dirichlet concentration parameters and random seeds. This facilitates the assessment of the stability of FL performance in various client data setups instead of a fixed division.

E. Environment

All experiments were run in Google Colab on an NVIDIA Tesla T4 GPU (16GB VRAM). Random seeds were fixed for reproducibility.

TABLE I
PERFORMANCE COMPARISON OF TRANSFORMER BACKBONES UNDER FEDERATED SETTINGS

Model	Fed IID Acc	Fed IID F1	Fed Non-IID Acc	Fed Non-IID F1
BERT-base	0.7801	0.8429	0.7387	0.7625
DistilBERT-base	0.7819	0.8446	0.7746	0.8339
RoBERTa-base	0.7819	0.8446	0.7820	0.8446

V. Results

A. Backbone Model Selection

BERT-base, DistilBERT-base, and RoBERTa-base were tested in order to determine model to use in the FL framework. All the models were tested on both IID and Non-IID data distribution, but there was differences in the federated environments where there is Non-IID distribution of client data. The outcomes mentioned here are an average of three runs that involve three clients and four rounds of communication.

All of the models gave similar outcomes under the IID condition, yet the Non-IID outcomes reveal some critical differences. BERT is the most degraded and DistilBERT is moderately stable. RoBERTa has almost the same IID and Non-IID performance, and it is more robust to heterogeneous client data. Since healthcare has applications that need reliability, RoBERTa-base is chosen to be the base model. RoBERTa was chosen as it offers better contextual information and better detection results than lightweight models, which is why it is appropriate regardless of its moderate computational cost.

B. Model Performance Evaluation

1) Centralized: The initial training of RoBERTa-base was conducted on a centralized basis to determine an upper limit of performance. The model has an accuracy of 0.96, F1-score of 0.9669, and an AUC of 0.9958 which proves to be effective in the event that all the data are centrally available.

2) Federated IID: In the case of IID federated training, when the clients are given equal data partitions, the model has the closest performance as centralized with an accuracy of 0.9572 and an F1-score of 0.9644. This implies that there is little degradation in data distributions that are homogeneous.

3) Federated Non-IID: In real-life Non-IID conditions where the distribution of client data is uneven, the first communication round has a small drop in performance, and then the performance gradually improves, which finally reaches a final accuracy of 0.9545 and an F1-score of 0.9622.

To further test, FL was tested on three Dirichlet-based NonIID splits with concentration parameters of alpha = 1.0, 0.5, and 0.3 which are mild, moderate, and severe heterogeneity. Performance is also similar across all configurations, meaning that the model shows limited sensitivity to client partitioning.

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These settings ROC analysis also supports the stable ability to differentiate when the data heterogeneity changes.The model was able to achieve similar performance across multiple rounds, indicating stable and reliable performance.

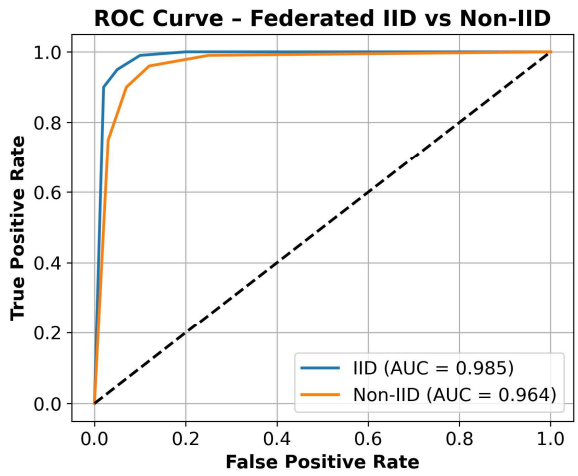


Fig. 2. ROC curves for IID and Non-IID federated settings.

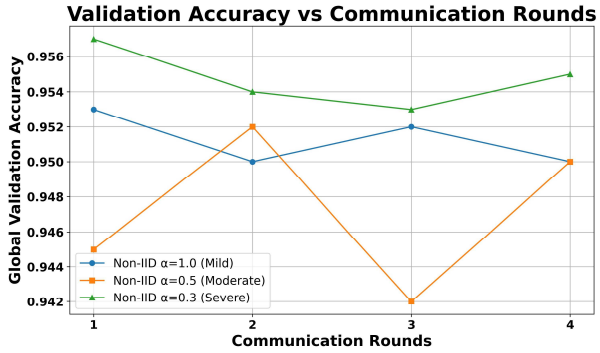


Fig. 3. Global validation accuracy across communication rounds for Non-IID client configurations (α =1.0,0.5,0.3).

C. Federated Training Behavior Analysis

To evaluate the stability and convergence of the federated model, we analyzed ROC performance, global validation accuracy, and average client training loss across communication rounds.

1) ROC Curve Comparison (IID vs. Non-IID): The performance of both IID and Non-IID setups are good in terms of ROC with AUC scores of 0.996 and 0.984 respectively. There is a slight degradation in Non-IID conditions, the model can still correctly detect the benign and malicious prompts.

2) Global Validation Accuracy over Communication Rounds: In all Non-IID settings, there is a slight loss of accuracy in the model in the initial rounds because of the data distributions being skewed but quickly approaches the same performance and remains stable. This tendency shows the strength of the proposed FL framework to client heterogeneity.

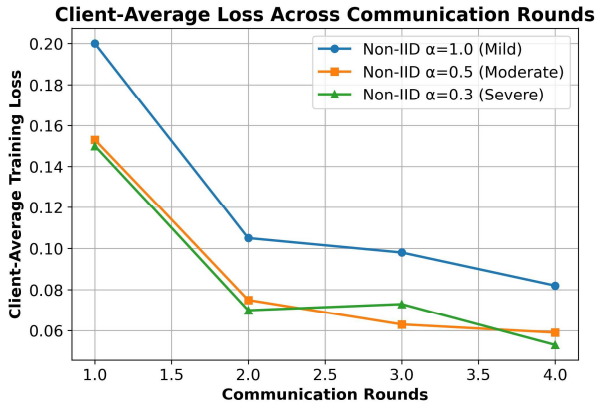


Fig. 4. Client-average training loss across communication rounds under different Non-IID settings (α =1.0,0.5,0.3).

3) Client-Average Loss Trend: The training loss decreases progressively throughout communication rounds of all NonIID configurations. Even though heterogeneity is increased, initial loss is always reduced and convergence is observed, which implies the presence of stable and effective federated optimization in realistic distributions of healthcare data.

VI. Discussion

While comparing with a centralized prompt injection detection approach[18], decentralization does not come at the cost of detection performance. Although centralized ensemblebased approaches are characterized by high accuracy and the ability to discriminate, the suggested RoBERTa-based federated model is equally accurate and slightly better in its F1-score and AUC in both IID and Non-IID conditions.

The federated model is stable at different degrees of client data heterogeneity, which implies strong resistance to skewed and fragmented healthcare data distributions. An important aspect of this approach is that we are not storing the raw patient data in the central server, which reduces the risk of data leakage and poisoning. However, practical deployment of FL can face challenges such as communication overhead from large model parameters, increased latency during aggregation rounds, and coordination difficulties between distributed clients. Despite these limitations, FL offers a good balance between privacy and effective detection, making it a suitable approach for privacy-preserving healthcare chatbot systems.

TABLE II
PERFORMANCE OF THE PROPOSED FEDERATED ROBERTA MODEL UNDER IID AND NON-IID SETTINGS

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Setting	Accuracy	AUC
Centralized [18]	92.24	0.994
Federated IID	95.72	0.996
Non-IID (α =1.0)	95.45	0.984
Non-IID (α =0.5)	95.38	0.995
Non-IID (α =0.3)	95.29	0.994

VII. LIMITATIONS

The framework has shown a promising performance but there are several limitations. The RoBERTa model introduces additional computational and communication overhead, which can affect deployment in low-resource settings. The evaluation was limited only to a single dataset and controlled experimental conditions, which may not completely reflect real-world federated healthcare scenarios.

VIII. CONCLUSION

This paper has looked into prompt injection defense in healthcare chatbots using a RoBERTa-based model under centralized, IID federated, and Non-IID FL conditions. From the experiments held, it clearly shows that the performance is comparable to centralized approach while remaining stable under heterogeneous client data distributions.

Since the framework does not store sensitive prompts on a central server it reduces privacy risks while also maintaining effective detection of malicious inputs. The findings show that FL can provide a practical balance between privacy preservation and prompt injection detection in healthcare environments. Future work will focus on evaluating the framework with larger datasets, more number of participating clients, and additional defense mechanisms to improve robustness against evolving prompt injection attacks. Improvements can also include integrating secure aggregation and differential privacy techniques to strengthen data protection in healthcare AI systems while supporting Sustainable Development Goals such as SDG 3, SDG 9, and SDG 16.

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